



Neither invented nor shared here: The impact and management of attitudes for the adoption of open innovation practices

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ABSTRACT

Despite the massive interest in open innovation, limited attention has been expressed concerning the intra-organizational challenges in implementing it. An exemplary issue is the unwillingness of employees to undertake extra-organizational knowledge transactions in the form of negative attitudes against the sourcing of external knowledge (the not-invented-here (NIH) syndrome) and against the external exploitation of knowledge assets (the not-shared-here (NSH) syndrome). Using survey data collected from 331 firms, this article empirically assesses the theoretical assertion that the NIH and NSH syndromes have negative impacts on the adoption of inbound and outbound open innovation. Furthermore, it investigates how their effects can be reduced through competence-building programs based on the training of employees. By focusing on two attitudinal antecedents to openness, the findings offer an explanation for the problems that firms face in benefiting from inflows and outflows of knowledge and possible guidance as to how managers can disengage such attitudes.

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1. Introduction

“The management team [of our partner company] saw us as being incompetent and, in hindsight, did not recognize that it was due to their own employees unwillingness to outsource. The employees who felt the assignment belonged to their R&D department wouldn't let go of it. In a clever way, they managed to ensure that a number of things did not work out – and that can easily be done.” quote from the CEO of an engineering consultancy.

This opening vignette illustrates great resistance from the employees of a partner company whose management had decided to outsource part of its R&D activities. The vignette illustrates the importance of employees' attitudes for the successful implementation of open forms of organizing for innovation. It epitomizes an issue many organizations currently face, as innovations increasingly become leveraged on the basis of collaborative practices carried with external partners (Ahuja et al., 2008; Cassiman and Veugelers, 2006). This trend, known as ‘open innovation’ (OI) (Chesbrough, 2003), has been increasingly recognized (Gassman et al., 2010; Huizingh, 2011; Knudsen and Mortensen, 2011; Van de Vrande et al., 2009) and acclaimed for its potential to lead to improved innovative output and firm performance (Laursen and Salter, 2006). As a result, the recommendation that companies should become more open to support innovativeness has prevailed in the discourse of academia and the business press for the past

decade (Chesbrough and Garman, 2009; Foss et al., 2011). Yet, as the experience of the engineering consultancy suggests, the way employees feel about such initiatives is decisive. Thus, in face of the increasing awareness about openness, are organizations and employees willing to receive and share knowledge with others?

The evidence available thus far is relatively scarce. Prior studies on employee attitudes to knowledge are based on anecdotal accounts or qualitative studies, with the notable exception of Herzog and Leker (2010). As a result, there is no consensus in the literature about which attitudinal tendencies prevail, and particularly how they affect the implementation of new forms of organizing for innovation such as open innovation. Besides, limited knowledge exists about how to foster the mindsets and the attitudes necessary for a company to engage in knowledge transactions in collaboration with external partners (Gassman et al., 2010; West and Borges, in press).

Empirical evidence about employee attitudes to knowledge is not only scarce, but also inconsistent, as it suggests contradictory results. While most studies have found that employees tend to be unwilling to collaborate, i.e. negative attitudes to knowledge sharing predominate (Herzog and Leker, 2010; Michailova and Husted, 2003; Mortara and Minshall, 2011; Reitzig and Sorenson, 2010), others have documented the existence of overly positive tendencies to knowledge insourcing (Menon and Pfeffer, 2003; Menon et al., 2006). Hence, although attitudes to knowledge have long been theoretically described as individual- or group-level phenomena related to the technical performance of organizations (Katz and Allen, 1982), little and inconsistent empirical evidence exists around them.

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Since employees carry a substantial responsibility for the actual adoption of new innovation management principles like open innovation, understanding the consequences and the managerial drivers of their attitudes to knowledge is of crucial importance. If the individual employee or group of employees is misaligned with top management in their desire to implement new practices, managers face significant challenges, which may threaten the very survival of companies (Huston and Sakkab, 2006; Lucas and Goh, 2009). Moreover, when confronted with undesirable attitudes, managers need to know how to mitigate them. This is important since attitudes are individual predispositions to respond to given objects which ultimately affect the actual behavior of people (Fishbein and Ajzen, 1975). Nonetheless, their effect can be diminished or strengthened depending on the context in which they operate (Crano and Prislin, 2006).

For the adoption of open innovation practices in particular, two of the most significant attitudes are the negative attitude towards the utilization of external knowledge (i.e. the not-invented-here (NIH) syndrome) and the negative attitude against the external exploitation of knowledge assets (i.e. the not-shared-here or Not-sold-here (NSH) syndrome) (Chesbrough, 2003; Katz and Allen, 1982; Lichtenthaler and Ernst, 2006), since they are related to the two dimensions of the openness concept. On the one hand, inbound open innovation corresponds to the outside-in dimension, which encompasses practices related to the absorption of external knowledge, e.g. through in-licensing or strategic partnerships. On the other, outbound open innovation corresponds to the inside-out dimension which encompasses the active use or commercialization of knowledge outside a firm's organizational boundaries, e.g. throughout-licensing or free revealing of innovations (Dahlander and Gann, 2010). This article investigates the theoretical assertion that *the NIH and NSH syndromes have negative impacts on the adoption of respectively the inbound and the outbound approaches to open innovation*.

Following the proposition above, this article disentangles the relationship between attitudes and openness in addressing the following research questions: *are employees' attitudes to knowledge related to open innovation practices? How can management mitigate the effects of negative attitudes to knowledge?* The article intends not only to empirically test the consequence of attitudes to the adoption of open innovation (with the view of resolving the current inconsistencies in the literature), but also to provide further theoretical ground to this relationship by drawing on socio-psychological research. In addition, it develops theory about potential managerial moderators of the open innovation-attitude link by analyzing organizational procedures that may reduce the effect of undesirable attitudes, in particular those related to employee training programs. Furthermore, a comprehensive list of open innovation practices is composed with the aim of capturing the richness of the construct.

The article proceeds as follows. The following section addresses the theoretical background of attitudes. Based on these theoretical insights, three main hypotheses are developed in Section 3. The empirical work is described in the Section 4, followed by a presentation of the results in Section 5. The empirical results then lead to a full discussion of the findings in Section 6. The article concludes in Section 7 with implications for theory and practice and with suggestions for future research.

2. Attitudes to knowledge and innovation

2.1. The NIH and NSH syndromes as key attitudes for innovation

The NIH and NSH syndromes have been identified as crucial attitudes to knowledge in the context of open innovation

(Chesbrough et al., 2006; Huizingh, 2011). This is because the implementation of open innovation practices rests on an initial valuation of outside competence and know-how by the management, founded on the attitudes of employees and of management, and weighted against the organizations ability to fully exploit purposive inflows and outflows of knowledge. The recent studies by Van de Vrande et al. (2009), Mortara and Minshall (2011) provide evidence to this point, showing that the organizational and cultural issues which arise when companies start to interact with external partners were the main barrier to the implementation of open innovation strategies both in small and medium-sized firms and in large multinationals. Teirlinck and Spithoven (2013) and Pedrosa et al. (2013) add to this debate by evidencing the role of R&D personnel qualifications and managers' characteristics in the processing of external knowledge.

Although interrelated, the NIH and NSH syndromes constitute conceptually distinguishable constructs, as they are directed towards different objects, namely external knowledge acquisition (from outside-in) and external exploitation of knowledge (from inside-out). The first relates to a piece of knowledge that has been developed outside the organization, but which may possibly be appropriated and applied internally (Katz and Allen, 1982). Conversely, the latter refers to a piece of knowledge that has been developed inside the organization, but which may be applied by others outside (Chesbrough et al., 2006). Both attitudes may nevertheless constitute a misalignment between the intentions of top management and the behavior of involved employees, since they create unfavorable perceptions of the value of outside competencies and know-how, lending support only to internal development and application of ideas and technologies.

Despite being widely mentioned in the literature (Chesbrough et al., 2006; Lichtenthaler and Ernst, 2006; Michailova and Husted, 2003), few studies have attempted to assess attitudes to knowledge directly. As Katz and Allen (1982) first introduced the term 'Not-invented-here syndrome,' on the basis of communication data to explain the decrease in performance of R&D teams over time, subsequent research has to a large extent followed this tradition. Studies such as the ones of Reinholt et al. (2011) and Hansen et al. (2005) use data on intra-organizational networks to argue that an overemphasis on within-team relations and communication can be seen as manifestations of the NIH syndrome. While acknowledging the valuable contribution of these studies, this article adopts a socio-psychological lens that highlights the attitude-theoretic fundamentals of the constructs. Put differently, the NIH and NSH syndromes are conceptualized as attitudes at the individual level that can be observed among employees prior to the engagement in innovation-related activities and that can, therefore, be directly examined.

2.2. What are attitudes and why do they matter?

Given the focus of our research on the attitudinal characteristics of the NIH and NSH syndromes, we draw on socio-psychology research to discuss in greater depth the nature of these constructs. All attitudes are social in the sense that they develop, function and change within a social context. Even if expressed in individualistic terms, they are causally connected with a person's social relations and thus, with the dynamics of group processes. Prior work has gathered considerable evidence showing that attitudes are to a large extent driven by social and group influences (Prislin and Wood, 2005). In the context of firms, this implies that attitudes are deeply embedded in an organizations' culture, together with norms, values, artifacts and other social principles. More specifically, attitudes to knowledge constitute a key part of an organization's innovation culture (Herzog and Leker, 2010). Nevertheless, attitudes are a central part of human individuality serving to express values,

self-image, sense-making and thereby facilitate adaptation to the environment (Ajzen, 2001; Albarracín et al., 2005). They represent a judgment, or in more formal terms, “a psychological tendency to evaluate an identifiable object with some degree of favor and disfavor” (Spithoven et al., 2010: 745). Once attitudes integrate cognitions (beliefs) and affects (emotions) experienced in relation to an object, they shape what people see, hear, think and do (Bohner and Dickel, 2011). They represent, in this way, a “*learned predisposition to respond in a consistently favorable or unfavorable manner with respect to a given object*” (Fishbein and Ajzen, 1975). In general, three aspects are important for our understanding and interpretation of the effects of attitudes, which are that attitudes are *learned*, may be *favorable or unfavorable* and represent a *predisposition*. Accordingly, an employee does not alter his attitudes according to a situation, but acts according to the predisposition. Furthermore, the predisposition may be influenced or re-shaped as an effect of learning efforts, which is particularly interesting if the response is unfavorable (e.g. for a firm).

Attitudes are evidently important for organizations and managers as they guide *information processing* and influence *behavior* (Bohner and Dickel, 2011). Information processing is affected by attitudes to the extent that they function as selective filters to the information people expose themselves to. People predominantly seek out and pay close attention to new information that confirms their existing attitudes, typically attributing to it higher quality and other virtues that can reinforce their selection criteria. In this way, attitudes influence not only the attention to certain attitude objects, but also the interpretation, judgment and memory (Bohner and Wänke, 2002). In fact, the preference for attitude-congruent information (also known as congenial information) is a consistent finding within the socio-psychology field, indicating that attitudes do have a biasing effect in the handling of information (Bohner and Dickel, 2011). Besides, attitudes are key precursors of behavior or, more precisely, intentions to behave (Ajzen, 2001; Crano and Prislin, 2006; Glasman and Albarracín, 2006). Consequently, if a firm seeks to implement new ways of organizing innovative activities such as the open innovation approach, attitudes may, although not intentionally, play a dominant role either legitimizing or hindering such new initiatives in being adopted and implemented. Studies also suggest increased attention should be given to the emotional process when implementing innovation (Choi et al., 2011). Attitudes can constitute either important micro-foundations of or major barriers to the development of organizational capabilities at the firm level (Chesbrough et al., 2006).

3. Hypotheses

3.1. The consequences of attitudes to knowledge for open innovation

3.1.1. The NIH syndrome and inbound open innovation

The NIH syndrome is expected to shape both the way in which employees look for new information (e.g. screening out those of outside origin when they challenge their attitudes) and the way in which they assess it, eventually leading to the rejection of external knowledge due to its source and not based on its content per se (in terms of the feasibility, quality, usefulness and other related rational criteria). This results in what has been termed “attitude selectivity effects”, which influences the attention, elaboration and memory of new information. Such effects may be either strategic, conscious (i.e. actively seeking solutions exclusively with colleagues inside the organization) or automatic (i.e. involuntarily forgetting ideas received from external partners) (Bohner and Wänke, 2002). In this way, protective attitudes are expected to create judgmental biases regarding the valuation of outside knowledge throughout

the search and the application stages of inbound open innovation, which in turn favors internal production and leads to resistance to technological knowledge from the out-group (Lichtenthaler and Ernst, 2006; Lucas and Goh, 2009; Reitzig and Sorenson, 2010).

Furthermore, employees often feel threatened by new ideas from others and therefore tend to devalue them in an attempt to promote internal projects and in-house capabilities. In other words, a kind of knowledge provincialism may emerge inside companies with NIH tendencies (Reitzig and Sorenson, 2010), where it is considered more legitimate and prestigious to create new knowledge in-house rather than reuse knowledge invented elsewhere (Michailova and Husted, 2003). By affecting information processing activities related to external knowledge, the NIH syndrome is expected to affect the actual behavior, that is, the extent to which employees truthfully scan the environment for new technologies, collect industry information, approach external institutions and establish collaborations with partners or lead users to acquire knowledge. An employee is thus less likely to actively try to insource external knowledge if he/she is unfavorable to it. Therefore the article posits the following:

H1. The higher the negative attitude to external knowledge acquisition (i.e. NIH-syndrome), the lower the level of inbound open innovation practices.

3.1.2. The NSH syndrome and outbound open innovation

In a similar vein, the adoption of outbound open innovation practices is a highly complex activity. This is due not only to the intangible and idiosyncratic nature of disembodied knowledge (in the form of technologies, patents, know-how and ideas) for which severe market imperfections exist (Fosfuri, 2006), but also to the variety of related sub-tasks, ranging from technical to marketing and legal (Guilhon et al., 2004). It thus involves the establishment of a myriad of processes, structures and activities to support an active identification of potential receivers of knowledge, as well as the handling of spontaneous demands from external partners (Chesbrough, 2003). Yet, if NSH tendencies exist in this context, employees feel that if the knowledge or technology cannot be exploited in own products or markets, it should not be exploited at all. This emotional bias affects the way in which employees transmit related information, engage in collaboration and negotiate contracts with partners, making them refrain from sharing knowledge even if it may not be useful for internal purposes. Hence, an employee who is not in favor of exploiting ideas outside the organizational boundaries would (i) easily miss out potential opportunities, e.g. in out-licensing of technologies, as he has attention and memory involuntarily drawn away from this activity; (ii) be prepared to respond negatively to any novel situation he encodes as an opportunity for external knowledge sharing; and (iii) make negative inferences and judgments about it (i.e. considering that giving away potential value will invariably dilute the organization's core competencies). This attitude thus induces the evaluation of a situation of inter-firm knowledge sharing as harmful, independently of the context in question, which may or may not be bad for the organization.

In addition to influencing the conscious and unconscious perception of attitude-congruent information, the NSH syndrome affects in a more strategic fashion behavioral decisions too. Put differently, individuals who present this attitudinal trait are less inclined to engage themselves in or execute outbound-related activities and may even actively seek to limit them, motivated by the belief that they do not want to sell out relevant in-house capabilities. As a result, we expect that negative attitudes to the exploitation of knowledge and technologies outside the walls and normal channels of the company (e.g. via out-licensing agreements,

the setting up of ventures or free revealing) hamper the adoption of outbound OI. Therefore the article posits the following:

H2. The higher the negative attitude to external knowledge sharing (i.e. NSH syndrome), the lower the level of outbound open innovation practices.

3.2. Influencing the impact of attitudes through employee training

Notwithstanding the expected negative consequences of these attitudes, their effects are likely to be subject to managerial discretion. Given that attitudes are learned predispositions, a way to diminish their negative consequences is to reduce or even invert predispositions by applying positive learning mechanisms that may weaken them. One option for limiting the effect of negative predispositions is to apply human resource development tools such as training, since they can improve the knowledge, skills and the abilities of the employees, thereby increasing the motivation and commitment to the tasks of the organization (Ji et al., 2012). Training is one way in which employee on-the-job learning can occur and is defined as “the systematic acquisition and development of the knowledge, skills, and attitudes required by employees to adequately perform a task or job or to improve performance in the job environment” (Tharenou et al., 2007: 252).

Training offered within the organization is associated not only with functional benefits, in the sense that employees acquire key skills for fulfilling their job responsibilities, but also with intangible benefits, in the sense that employees become familiarized with the desirable values and attitudes of the organization (Ehrhardt et al., 2011). This is because training programs elicit and reinforce the attitudes and behaviors required by the organization's strategy (Tharenou et al., 2007), thus weakening the predisposition of employees to act according to individual preferences. They function, in this way, as socialization forums, where desired role behaviors are communicated and promoted, resulting in social support (and pressure) to conform to them. The role of social influence is substantiated in the findings of socio-psychology research too, which indicate that social influence has key evaluative implications, in the sense that positions are valued according to the level of support they receive. The degree of social influence bias the processing of attitude-relevant information, which in turn influences the likelihood of individuals to act on their attitudes (Crano and Prislin, 2006).

In addition to competence development tools, employee training often includes incentive systems too. Promotions, pay raises and performance evaluations are some of the incentives used to further induce the implementation of intended work practices. Incentive systems are crucial in the context of attitudes, once they

influence the strength of the causal relationship between attitudes and intentions to behave and, consequently, actual behavior. If incentives are strong enough, they may motivate employees to circumvent own attitudes (when incentives are not in line with existing attitudes). In consequence, a better alignment between the attitudes of the staff and of management is one of the key expected outcomes of training. Put differently, training programs contribute to legitimize the desired attitudes of an organization; thereby restraining the extent to which individual attitudes that do not comply with collective attitudes come to affect actual work practices, i.e. minimizing agency issues.

There is indeed a growing body of evidence in the field of human resource management recognizing that, as a development activity, training provides a useful vehicle for modifying employee attitudes, either via direct or indirect effects (Ehrhardt et al., 2011; Frisque and Kolb, 2008; Ji et al., 2012). In particular, empirical research focusing on attitudes to work has found that, in firms with greater training, employees jointly report higher levels of job satisfaction (Tharenou et al., 2007). Overall, the benefits of training are increased firm performance with respect to work climate, quality of customer service and organizational effectiveness (Ji et al., 2012; Tharenou et al., 2007).

This line of reasoning also finds support within the dynamic capabilities perspective, which states that building competitive advantage through innovation is a question of supporting organizational processes inside the firm (Teece et al., 1997). According to this perspective, training can be viewed as an investment in human capital (Tharenou et al., 2007) aimed at building the “special know-how” needed to sense the environment, seize opportunities and reconfigure assets (Teece, 2007). That is, training is viewed as a mechanism of upgrading skills, routines and practices and of renewing resources and capabilities, which enable organizations to become more agile and flexible. As the development of new resources and capabilities is not solely anchored on firm-level processes, but also on key individual level foundations (Felin and Foss, 2005; Rothaermel and Hess, 2007), it is expected to influence the values and beliefs that lie in employee attitudes.

Hence, contributions from the human resource, socio-psychology and dynamic capabilities literature support the argument that employee training can weaken the effects of attitudes. Therefore the article posits the following:

H3a. Employee training positively moderates the effect of the NIH syndrome on inbound open innovation practices.

H3b. Employee training positively moderates the effect of the NSH syndrome on outbound open innovation practices.

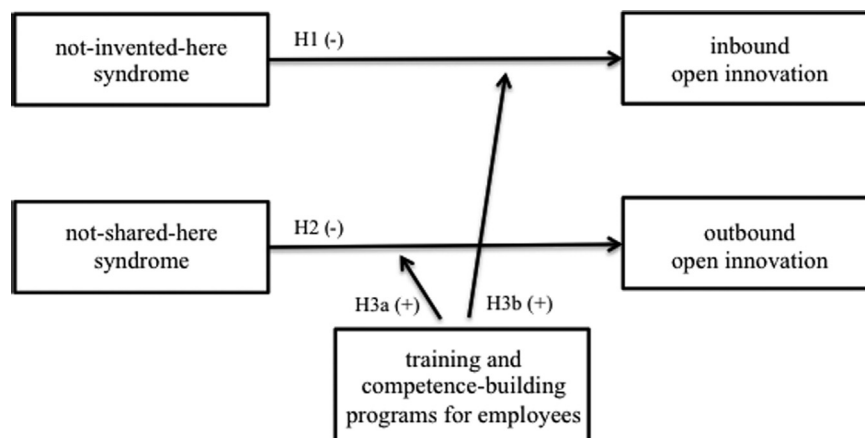


Fig. 1. Model overview and formulated hypotheses.

The theoretical model and the developed hypotheses are shown in Fig. 1.

4. Methodology

4.1. Questionnaire design and data collection

Our article gathered survey data on private manufacturing firms in Denmark. Data was collected using an online instrument during the period from August to November 2010. The questionnaire was developed by the research team during the spring 2010 and was pre-tested by experts (researchers from universities) and by two companies. These pre-tests led to the refinement of item wording and of the survey structure. To the extent possible, the questions were based on previous studies of open innovation and human resource practices. Yet, since some of the scales were not available in the literature, they were developed for the purpose of this research. The survey was formulated in Danish since the respondents had this as their native language. The scales from the literature were thus translated from English into Danish.

The population was drawn from a Danish nationwide electronic database (NED), which is regarded the most comprehensive database of companies available. The population includes small and medium-sized enterprises (SMEs) with 5–499 employees belonging to the R&D (NACE 72) and manufacturing industries (NACE 10–37). Applying these selection criteria based on firm size and industry affiliation resulted in the identification of 4445 firms, which were approached to provide responses to the survey questionnaire. Prior to the administration of the electronic survey, all firms were contacted by telephone so that the study could be presented and the respondents could be identified. The phone calls were conducted by specially trained research assistants and ensured that the person responsible for innovation, either the R&D or the innovation manager, was identified and willing to receive the questionnaire. R&D managers were considered the most suited informants for our survey, because they are the most knowledgeable about the innovation-related aspects of companies, which constitute the main focus of our questions. However, in the case of small businesses without these job titles/functions, the business owner or general manager was asked to participate. In a few cases, only the general company email was obtained, from which the company distributed the questionnaire internally. In order to verify sufficient proficiency, the respondents were requested to have more than three years of work experience in the company to be included in our study. In addition, companies were screened out if they had no innovation-related activities in the three years prior to data collection, as this was considered the maximum timespan respondents could recall when responding to the questionnaire. The phone interviews resulted in 1051 company e-mails to which the questionnaires were sent. After two reminders, a total of 345 responses were returned. After screening for inconsistent answers, there were 331 usable questionnaires, for a response rate of 7.44% (of the original 4445 firms, and the response rate is 31.49% of those that accepted to participate in the survey – see Table 1).

Subsequently, the data was tested for representativeness comparing the population with the firms that accepted and the ones that ended up participating in the survey. These groups of firms were tested on the basis of industry classification (NACE code 1 digit), size (6 size groups), and regional distribution (5 administrative regions of Denmark). As the comparisons did not reveal any significant differences between any of the three groups (population, accepts and participation), we can accept that the sample is representative of the population.

Table 1
Response rates for the survey.

Response rates	Population		Number of accepts	
	Number	Percentage	Number	Percentage
Base	4445	100	1051	100
Before 1st reminder	145	3.3	145	13.8
Before 2 nd reminder	250	5.6	250	23.8
Total	345	7.8	345	32.8
Usable answers	331	7.44	331	31.49

Note: 331 cases were used in the final analyses, because 7 cases were deleted due to inconsistent answers and missing responses on key questions and 17 cases were deleted because companies reported firm size beyond our target (5–499 employees).

4.2. Measurement

4.2.1. Open innovation

Open innovation is measured adopting a practice perspective, that is, according to a set of innovation practices related to the inbound and outbound dimensions, such as purchasing of R&D work and actively participating in other companies' innovation process. The aim is to capture the multifaceted nature of the concept by including both the directional dimensions, as well as to include practices that transcend the relationship approach (Laursen and Salter, 2006). The selection of practices was generated from a literature review. After screening research addressing open innovation topics (Bahemia and Squire, 2010; Chesbrough and Garman, 2009; Van de Vrande et al., 2009), a comprehensive list of 11 practices was identified, with the aim of building an extensive rather than a compressed list of practices. The average number of practices used for all companies is 6.45 (mode=7), while firms that have introduced new products on the market in the last 3 years (6.84) use a significantly higher number of practices compared to the firms that did not introduce a new product in the last three years (5.14) (Esbjerg et al. (2012)). The practices used represent opportunities for either acquiring and or sharing knowledge by offering information, data or know-how to the firm. Whether this in reality is transformed into knowledge depends on the further handling of the information in the firm. The list of practices includes more inbound than outbound practices (respectively 8 inbound and 3 outbound), which is a reflection of the hitherto focus on this dimension in the literature and of the fact that firms may be more inbound oriented. Table 2 provides an overview of the practices investigated, as well as the share of firms in the sample using them. The most frequently used practice is searching the internet (86%) and the most frequent outbound is the knowledge provision (54%).

The OI practices were measured using binary variables (i.e. 1: used/ 0: not used). Inbound OI and outbound OI variables were then coded as the sum of the number of respectively inbound and outbound practices used. The underlying assumption is that firms implementing more practices are more open to innovation. The mean number of implemented inbound practices is 4.28 (st. dev. 1.62) with a maximum of seven practices implemented ($n=14$ firms; 4.2%), out of the possible eight practices. Similarly, the mean number of outbound practices is less than one (mean=0.69 and st. dev.=0.61), while the highest number of implemented practices (two) is only achieved by 25 firms (7.6%). Thus, inbound is more often adopted than the outbound OI practices by firms in our sample, reflected both in the number of identified practices (8 inbound vs. 3 outbound) and the number of firms adopting 0 practices (3.3% implementing 0 inbound; 39.1% adopting 0 outbound practices).

Table 2
The use of the open innovation practices among Danish SMEs.

Construct	Practices	Yes	%
<i>Inbound open innovation practices</i>	Used the internet to search for new trends or technology	284	86
	Reading technical magazines	275	83
	Used information from trade organizations	257	78
	Participated in innovation related fairs or shows	229	69
	Purchased R&D work from others	197	60
	Purchased licenses, patents or know-how	83	25
	Worked with lead users	74	22
<i>Outbound open innovation practices</i>	Used innovation brokers	17	5
	Actively participated in other's innovation projects	178	54
	Sold patents, licenses or know-how	27	8
	Made own innovations available to others for free	17	5

N=331, Question: Please indicate which of the following methods were used by your company within the past 3 years.

To analyze these practices as dependent variables, they were compiled into two categorical variables. This was an important step to operationalize our estimations, as the number of original categories did not follow a normal distribution. We re-coded each dependent variable according to the level of use of the practices, i. e. low, medium or high. The following categories for inbound OI (low: 0–2 practices; medium: 3–5 practices; high: 6–7 practices) and outbound OI (low: 0 practice; medium: 1 practice; high: 2 practices) were adopted. Several versions of the variables have been tested with limited effect on the results.

4.2.2. Not-invented-here and not-shared-here syndromes

The multi-item scales used to assess the NIH and NSH syndromes were designed to capture the overall level of employees' attitudes to knowledge on a 7-point Likert scale. Yet, as the objective was to collect a large-scale dataset, allowing for inter-firm comparisons, the simultaneous collection of data from all members of the staff in each firm was not feasible, although recommendable. Rather, R&D managers were asked to assess employees' general evaluation of the attitude objects – external knowledge acquisition and external exploitation of knowledge (Bohner and Dickel, 2011; Bohner and Wänke, 2002). The intention was to obtain a group-level proxy of attitudes for each company (i.e. the organizational average pertaining to the R&D and innovation tasks), and not an account of individual differences at the intra-organizational level. For this purpose, R&D managers were deemed appropriate informants to make confident evaluations of these attitudes, since they are most knowledgeable about the diverse aspects within their organizations that either relate directly to or influence the decisions regarding innovative activities. This is especially the case for SMEs, where managers typically exercise a generalized job function, which provides them with close contact with all parts of the organization (Florén and Tell, 2004).

Drawing on socio-psychology research, the items used to assess the NIH and NSH syndromes are instruments for direct and indirect attitude measurements, both being widely used in both academic and applied research (Bohner and Wänke, 2002). While some items were phrased in a more straightforward and objective way (items 1 and 2 – see Appendix A, Table A1), others were aimed to capture past behavior in external knowledge acquisition and external knowledge exploitation (items 3 and 4). On the one hand, the simplest way of assessing a person's attitude is to ask her. On the other, once attitudes are not directly observable, behavior is often used as a way of drawing inferences about attitudes (Ajzen, 2001; Albarracín et al., 2005). Both direct and indirect ways of asking about attitudes were used with the view of increasing the reliability of the scales.

In addition, with the view of increasing their validity, questions were developed as far as possible on the basis of existing scales. Specifically, the items pertaining to the NSH scale were adapted from Lichtenthaler and Ernst (2009) scale of 'organizational climate' that aimed at measuring employees attitudes to external commercialization of knowledge (including one addition of item 4 to capture the positive side of attitudes). Rather than adopting the exclusive view on the commercial side of knowledge provision (knowledge selling) as in the original scale (typical of out-licensing agreements), we rephrased items to focus more generally on knowledge sharing by including situations of free revealing or sharing without financial tightening. As suggested in the provider perspective (Tranekjer and Knudsen, 2012) and the broader inter-organizational relationship literature, the scale is designed to reach beyond pecuniary transactions and to involve less formalized collaboration patterns in reflecting the attitudes towards external knowledge provision. Because a similar survey instrument for the NIH syndrome could not be found in the literature, the NSH constructs were mirrored to reflect the attitudes towards knowledge acquisition.

To ensure construct validity, discriminant validity and reliability, the analyses included both exploratory and confirmatory factor analysis. The exploratory factor analysis (varimax rotation with Kaiser normalizations) with all the seven items resulted in a two higher-order factor solution, whose item loadings corresponded clearly to the intended constructs. There is thus evidence that the two attitudes are empirically distinct, which corroborated our theoretical expectations. Confirmatory factor analyses were then performed on each variable, and they further confirmed the validity of these constructs. The standardized factor loadings were all greater than 0.5 (see Appendix A, Table A1), as well as the average variance extracted for both variables (AVE=0.64 for the NIH syndrome; AVE=0.59 for the NSH syndrome). Besides, the AVE estimates for two factors are greater than the square of the correlation between the two factors (0.37) (Hair et al., 2009). The Cronbach Alpha levels were also acceptable at levels $\alpha=0.71$ (NIH syndrome) and $\alpha=0.76$ (NSH syndrome). The summarized variables were computed by calculating the arithmetic averages of their respective items so that they could be treated as observed indicators and then standardized. Finally, it is worth noting that companies in our sample presented higher levels of NSH tendencies (average: 4.09 on a 7-point scale) than of NIH tendencies (average: 2.79 on a 7-point scale).

4.2.3. Moderators

With the view of investigating the types of employee training that are central to diminish the effect of negative attitudes to knowledge, three moderator variables were applied to assess firms' efforts in

competence development. They were chosen because they entail various content forms and are intended to develop different skills and competences among employees, namely general abilities, innovative abilities and leadership skills. The three types of training chosen were the following: (1) training for professional competencies; (2) training for creativity and innovation; (3) the firm has nursed special talents in the firm e.g. by providing special training opportunities to selected employees. See [Table B1 in Appendix B](#) for the exact formulation of the items.

Following state-of-the-art research, training was operationalized as a content measure (i.e. according to the type of training provided) on the basis of a single-item variable. This is the most widely used form of scale to measure training ([Tharenou et al., 2007](#)). Each variable was thus measured as a binary variable (1=yes; 0=no). ‘Training for professional competencies’ (mean=0.79; st. dev. 0.40) relates to the deepening of already existing skills of the employee, e.g. by adding new skills in the employees’ field of expertise. It is not surprisingly the most common form of training utilized in the sample. ‘Training for innovation and creativity’ (mean=0.32; st. dev. 0.47) relates to the formation of specialized competencies. This form of training provides employees with enhanced awareness and understanding of the innovation process, as well as expands skills with regards to idea generation, concept development and entrepreneurial initiative ([Birdi, 2007](#)). It is expected to signal commitment to innovation from the part of the management, but also appreciation for the special role employees play in these activities. Finally, ‘nursing special talents’ (mean=0.35; st. dev. 0.48) is also a form of training related to the development of specialized skills, yet with particular focus on leadership skills. It refers to training programs aimed at “high potentials”, the promising employees an organization hand-picks because it believes that they have the requisites to handle responsibilities of greater scale and scope, i.e. become future leaders. Special talent programs typically include job rotation schemes, reward and incentives, as well as special terms for certain employees to e.g. become a mentee or with special opportunities with focus on empowerment ([Fernández-Aráoz et al., 2011](#); [Stahl et al., 2012](#)).

4.2.4. Control variables

In addition to the main variables, two types of control variables are included: (1) structural characteristics that include firm size, firm age and industry affiliation and (2) innovation-oriented variables including patent applications and time for creativity.

Firm size is measured by the natural logarithm of the total number of employees, as it is expected to impact positively the extent to which firms engage in open practices ([Huizingh, 2011](#)). The sample of firms contains mainly smaller companies, where 74.6% of the companies employ less than 50 employees. Firm age is calculated as the year of data collection (2010) minus the year of establishment, and then the natural logarithm of the number of years since the company foundation is used. The expectation is that older firms lack the flexibility needed to adopt openness ([Van De Vrande et al., 2010](#)). Industry affiliation is characterized according to the OECD classification of industries in terms of technology intensity (i.e. high-tech=1 and medium- and low-tech=0; low-tech=1 and high- and medium-tech=0). Hence, two dummies are coded and used. The majority of firms participating in the study are medium-tech firms, equaling 64.6%. This control accounts for the fact that the high technology intensity has been associated to higher levels of openness ([Chesbrough and Crowther, 2006](#)). Patent application is a dummy variable that accounts for the firms that have applied (once or several times) for protection of its innovations using patents (1=yes; 0=no) in the period 2007–2009. Intellectual property right has often been mentioned

in the literature as a key enabler of openness, as it increases the tradability of knowledge, offsetting some of the limitations of markets for technology ([Arora and Gambardella, 2010](#)). Finally, in order to control for the firm’s internal support of the innovative activities, the variable ‘time dedicated for creativity’ is added. It has been included because internal and external activities are considered complementary for the innovative activities of companies ([Hagerdoorn and Wang, 2012](#)), i.e. they are expected to positively impact openness.

4.2.5. Common method bias

As the data for our study is collected from a single key informant within each firm, who represents the view and the facts of the organization, the data may possibly be affected by common method bias. To limit concerns about this issue, we carried out post hoc statistical analysis using Harman’s single factor test. Following [Podsakoff et al. \(2003\)](#), an exploratory analysis (no rotation) was conducted with all underlying variables of our study. We found that the variables did not load to one single factor, nor lead to one general factor that accounted for the majority of the covariance in the dataset (the factor with highest variance explanation out of ten accounted for 14% of the total variance explained). Therefore, despite the limitations of Harman’s single factor test, it suggests that common method bias is not a serious issue in our analysis. The descriptive for the above variables and controls as well as correlations are presented in [Appendix C, Table C1](#).

5. Data analysis and results

Ordinal regression analyses were applied to test the hypotheses, since the dependent variables are discrete and ordered multinomial choice variables. As previously mentioned, the two dependent variables are each coded in three categories (low, medium and high use of practices) based on the number of practices applied. The test of moderation effects followed the approach and the recommendations presented by [Frazier et al. \(2004\)](#), who argue for a hierarchical regression approach. The use of the hierarchical regression approach implies a step-by-step inclusion of variables. The first step includes the control variables, the independent variables and the moderator variables. The second step includes the interaction term (i.e. moderator vs. independent variable), in addition to the other variables. As our model encompasses three moderators (‘training for professional competencies’, ‘training for creativity and innovation’ and ‘nursing of special talents’) for which the two steps were calculated, it led to the estimation of six models for each of dependent variable (respectively inbound and outbound open innovation). Thus, the article presents 12 regression models in total. The continuous independent variable is standardized to obtain a meaningful zero (here the mean). Since the moderators are dichotomous (0/1), the interaction effect (the moderator on the independent variable) either represents the full syndrome (NIH or NSH) or nothing.

5.1. The role of the NIH syndrome for inbound open innovation

[Table 3](#) presents the results of the ordinal regression analysis for inbound OI (models 1 to 6). In all six models, the innovation-oriented controls (time for creativity and application of patents) are highly significant. For the structural control variables, only a few are significant and to a limited extent (e.g. low-tech industry for models 2–6 and firm age for model 5). The control for firm size does not help in distinguishing the firms adopting inbound open innovation practices to different degrees.

Table 3

Results from ordinal regression analysis: consequences of attitudes to knowledge on the adoption of INBOUND open innovation practices.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Controls						
Time for creativity	1.18***	1.28***	1.18***	1.18***	1.29***	1.30***
Applied for patents	1.57***	1.51***	1.56***	1.51***	1.48***	1.61***
Firm size	0.39	0.46	0.42	0.49	0.39	0.54
Firm age	0.22	0.25	0.25	0.27	0.31*	0.26
High-tech industry	−0.63	−0.46	−0.43	−0.49	−0.37	−0.52
Low-tech industry	−0.59	−0.63*	−0.74*	−0.69*	−0.53**	−0.76**
Independent variables						
NIH syndrome	−0.40**	−0.87***	−0.49**	−0.70***	−0.40**	−0.18
NSH syndrome	−0.24	−0.25	−0.23	−0.22	−0.30*	−0.29*
Professional training	1.06***	0.89**				
Training for innovation			0.84**	0.86**		
Nursing of special talents					0.05	−0.16
Moderation effects						
NIH × professional training		0.62*				
NIH × training for innovation				0.53*		
NIH × nursing of special talents						−0.67**
Model statistics						
Model fit	0.000	0.000	0.000	0.000	0.000	0.000
Nagelkerke	0.352	0.363	0.366	0.377	0.337	0.353
McFadden	0.194	0.201	0.203	0.210	0.184	0.195
N	213	213	214	214	218	218

* Significant at 0.1.

** Significant at 0.05.

*** Significant at 0.01.

The model fits for all six models are good with acceptable fits for both Nagelkerke R^2 (ranging from 0.352 to 0.377) and McFadden R^2 (ranging from 0.194 to 0.210). Equally, for all models the test of parallel lines is insignificant, indicating that the impact from the dependent variable is the same on the probability of moving from one category to the next, irrespective of the considered category.

Models 1 and 2 present the models for 'professional training' (as independent variable and as a moderator of the NIH syndrome). Similarly, models 3 and 4 present the results for 'training for innovation and creativity', and models 5 and 6 present the results for 'nursing of special talents'.

The results with respect to the main effects reveal that the NIH syndrome significantly and negatively impacts the adoption of inbound OI in all models but model 6 (estimates range from −0.40 to −0.87, $p < 0.05$). For a continuous variable like NIH syndrome, a positive coefficient indicates that, as the values of the variable increase, the likelihood of larger scores increases and vice-versa. These results thus verify *hypothesis 1*. Moreover, the effect of the NSH syndrome on inbound OI was tested as to examine whether cross effects could be substantiated. NSH syndrome was found not to be significant in all models but model 6, indicating that the two attitudes indeed have distinguishable effects on the various dimensions of openness.

5.1.1. Moderation effects

Models 2, 4 and 6 respectively analyze the moderation effects of the three employee training variables. Our results reveal that 'professional training' and 'training for creativity and innovation' positively moderates the effect of NIH attitudes on the adoption of inbound OI ($\hat{\beta} = 0.62$, $p < 0.1$; $\hat{\beta} = 0.53$, $p < 0.1$ respectively), presenting a buffering interaction in the sense that they weaken the effect of the NIH syndrome (Frazier et al., 2004). It should be noted that the main effects are also significant ($\hat{\beta} = 1.06$, $p < 0.001$; $\hat{\beta} = 0.89$, $p < 0.05$ respectively), i.e. professional training and training for innovation and creativity constitute a partial moderator. Finally, for 'nursing

of special talents' the main effect is not significant ($\hat{\beta} = -0.16$, $p > 0.10$), but the moderating effect is negatively significant ($\hat{\beta} = -0.67$, $p < 0.05$), i.e. it is a full moderator with enhancing interaction effects. Taken together, these results partially verify *hypothesis 3a*, which posited a positive moderating effect of the variables related to employee training upon the relationship between the NIH syndrome and inbound OI.

5.2. The role of the NSH syndrome for outbound open innovation

Models 7–12 test in a similar fashion the hypotheses related to the main and moderating effects of the NSH syndrome on the adoption of outbound OI (Table 4).

Similar to the findings from models 1–6 on inbound open innovation, the control *time for creativity* is also significant ($p < 0.05$) for outbound open innovation, whereas application for patents is insignificant. These results indicate that prioritization of innovation work through time dedicated to creativity are important for the adoption of both dimensions of openness, possibly acting as a complement to the external activities. Contrary to the inbound OI results, the *age of the firm* is important for the adoption of the outbound OI practices ($0.05 < p < 0.1$). In all cases, the sign is negative, indicating that older firms are less prone to adopt outbound open innovation practices. The affiliation to *high-tech industry* is positive and significant in models 8–12, while the controls for *firm size* is not significant in any of these models.

The estimations of the main effects reveal that the NSH syndrome has a negative and significant impact on the adoption of outbound OI practices (estimates ranging from −0.49 to −0.87, $p < 0.05$). These results hence verify *hypothesis 2*. As it was seen in the models for inbound OI, the effect of the other syndrome, the NIH syndrome in this case, was tested. It was found not to be significant on outbound OI. This provides further corroboration to the fact that the two attitudes to knowledge are indeed distinct constructs with separate effects on openness.

Table 4

Results from ordinal regression analysis: consequences of attitudes to knowledge on the adoption of OUTBOUND open innovation practices.

Variables	Model 7	Model 8	Model 9	Model 10	Model 11	Model 12
Controls						
Time for creativity	0.79**	0.77**	0.84**	0.78**	0.85**	0.84**
Applied for patents	0.57	0.51	0.54	0.51	0.52	0.50
Firm size	−0.04	−0.15	−0.08	−0.06	−0.09	−0.03
Firm age	−0.41**	−0.39**	−0.36*	−0.36*	−0.32*	−0.31*
High-tech industry	0.88	0.98*	1.03*	0.96*	1.14**	1.19**
Low-tech industry	−0.17	−0.05	−0.11	−0.12	−0.07	−0.11
Independent variables						
NIH syndrome	−0.15	−0.14	−0.23	−0.22	−0.16	−0.16
NSH syndrome	−0.49**	−0.87**	−0.56***	−0.66***	−0.52***	−0.75***
Professional training	1.00**	0.89**				
Training for innovation			0.34	0.36		
Nursing of special talents					0.54*	0.54*
Moderation effects						
NSH × professional training		0.46				
NSH × training for innovation				0.36		
NSH × nursing of special talents						0.50*
Model statistics						
Model fit	0.000	0.000	0.000	0.000	0.000	0.000
Nagelkerke	0.262	0.268	0.252	0.257	0.247	0.260
McFadden	0.140	0.143	0.133	0.136	0.131	0.138
N	207	207	207	207	211	211

* Significant at 0.1.

** Significant at 0.05.

*** Significant at 0.01.

Table 5Results for *Hypotheses 3a* and *3b*: the moderation effects.

Variables	Inbound OI	Hypothesis 3a	Outbound OI	Hypothesis 3b
Professional training				
Main effect	Significant		Significant	
Moderation	Yes	✓	No	
Training for innovation and creativity				
Main effect	Significant		Not significant	
Moderation	Yes	✓	No	
Nursing of special talents				
Main effect	Not significant		Significant	
Moderation	Yes		Yes	✓

5.2.1. Moderation effects

The models 8, 10 and 12 test *hypothesis 3b*, which postulated positive moderation effects of employee training for the relationship between the NSH syndrome and outbound open innovation. The results differ with regards to the previous models (2, 4 and 6). Models 7 and 8 reveal that ‘professional training’ is itself a predictor for the adoption of outbound open innovation ($\hat{\beta} = 1.00$, $p < 0.05$), but it is not a moderator of the relationship between the NSH syndrome and outbound OI ($\hat{\beta} = 0.46$, $p > 0.1$). As model 10 reveals, ‘training for innovation and creativity’ is neither an explanatory variable ($\hat{\beta} = 0.34$, $p > 0.1$) nor a moderator ($\hat{\beta} = 0.36$, $p > 0.1$) of this relation. Finally, model 12 demonstrates that ‘nursing of special talents’ is directly related to the adoption of outbound practices ($\hat{\beta} = 0.54$, $p < 0.1$) and is also a moderator of the relationships between NSH and outbound OI ($\hat{\beta} = 0.50$, $p < 0.1$), presenting buffering interaction effects. Hence *hypothesis 3b* is partially confirmed with respect to ‘nursing of special talents’.

To provide a detailed overview of the moderation hypotheses, Table 5 summarizes related findings. These diverse moderation effects are discussed in greater depth in the following section.

5.3. Sensitivity analyses

For the main results presented above, it is primarily relevant to run sensitivity tests of the moderation variables. We tested both dummy coding (as reported here) and effects coding (as recommended by West et al., 1996: 10) for all moderators. These two coding systems did not result in different moderation effects, suggesting robustness in our results.

6. Discussion

This article has investigated the theoretical assertion that the NIH and NSH syndromes have negative impacts on the adoption of respectively the inbound and the outbound approaches to open innovation. The article consistently finds that NIH prevents the adoption of inbound open innovation practices, whereas NSH prevents the adoption of outbound open innovation.

Concerning the moderation effects, ‘professional training’ and ‘training for innovation and creativity’ were found to limit the effect

of the NIH syndrome upon inbound OI. Surprisingly, ‘nursing of special talents’ was found to enhance the impact of this attitude. This may be due to the fact that the development of leadership skills in special talent programs is rooted on the concepts of self-involvement and empowerment, which may further boost the tendency to undervalue knowledge coming from others. In addition, contrary to the expectations, the moderation effects of ‘professional training’ and ‘training for innovation and creativity’ were not found to be significant for the NSH syndrome on outbound open innovation. This may be due to the fact that the moderators do not capture the specificities of the particular type of training. We would expect that initiatives directly related to the problems and challenges faced in the single SME (e.g. customized training programs) have stronger effects than more standardized training programs oriented towards e.g. introducing employees to a set of general creativity techniques. Only ‘nursing of special talents’ appeared to be effective in limiting the impact of the NSH syndrome on outbound practices. The contrasting results of this form of training suggest that it has both negative and positive aspects. By extending the power of leaders of high-potentials (Stahl et al., 2012), special talent programs increase the confidence of these employees, leading self-sufficient attitudes in knowledge insourcing to be more impactful, while increasing the security in terms of knowledge sharing with external partners. Empowerment also allows employees the opportunity to demonstrate their capabilities, which seems to be important in restraining NSH tendencies, as they become better able to identify and evaluate the right outbound practices.

Interestingly, the various forms of employee training presented significant direct effects on the adoption of openness. ‘Professional training’ increased the extent of use of both dimensions of open innovation. This implies that the further development of skills and abilities within the field of expertise of employees, via training programs that provide them with depth of knowledge, support the application of inbound and outbound knowledge practices. In terms of the specialized forms of competence-development, ‘training for innovation and creativity’ was found to increase the level of adoption of inbound practices, whereas ‘nursing of special talents’ was found to increase the level of adoption of outbound practices. This may be explained by the fact that the development of leadership skills are important for the adoption of less institutionalized and less routinized practices such as the outbound innovation ones. They require the establishment of new process, new mindsets and structures for which the figure of a leader that can drive change and commitment in the organization is essential. In contrast, the insourcing of external knowledge (e.g. inbound OI) is based on generally more widespread practices that are likely to be supported in standard innovation training programs.

These findings related to the main effects of the moderators are in line with the human resource management literature, which has consistently found positive effects of employee training on firm performance (Ehrhardt et al., 2011; Ji et al., 2012). With respect to innovation outcomes, employee training has been considered effective too (Laursen and Foss, 2003; Shipton et al., 2006). Prior studies suggest that both firm capabilities and attitudes related to innovation can be developed through the training and development of employees (Ehrhardt et al., 2011; Jørgensen et al., 2010). In the same vein, the creativity literature emphasizes training as a means of encouraging employees to generate more ideas and to stay open-minded, as it has been found to lead to positive improvements in attitudes associated with divergent thinking (Birdi, 2007).

Employee training has a dual positive effect, which should encourage managers to focus more on such investments for innovation activities. On the one hand, in the role as a moderator, employee training limits or enhances the influence of employees’ protective tendencies to knowledge. On the other, in the role as

a direct driver, employee training provides employees with skills and competencies that support the adoption of open practices. Thus, employee training is one of the avenues management may take to get around a (perhaps too) high level of internal focus on innovation, legitimizing and creating incentives for external knowledge transactions.

7. Conclusions

We find that open innovation practices are related to employees’ attitudes to knowledge. Specifically, our results reveal that the level of negative attitudes to the acquisition and sharing of knowledge (the NIH and NSH syndromes) influences negatively and separately the extent of use of open innovation practices (the inbound and outbound dimensions respectively). In addition, we find that management can use specific types of training programs to diminish the impact of these attitudes. Professional training and training for innovation and creativity are effective against NIH tendencies, while individualized forms of training (the nursing of special talents) are effective against NSH tendencies.

7.1. Implications for theory and practice

The article contributes to the organization literature by testing and further developing theory about the role of attitudes to knowledge in innovation management. On the one hand, we have tested for a large sample of companies the theoretical assertion that the NIH and NSH syndromes have a negative impact on adoption of the open innovation approach. In this way, we could specify an important intra-organizational antecedent of openness, by indicating that employees’ attitudes indeed contribute to the understanding of the variance in the adoption of open innovation. This is a key theoretical contribution, in light of the current discussion about the relevance of these attitudes (in particular the NIH syndrome) as an organizational barrier to openness, based on qualitative data which suggested contradictory effects (Menon and Pfeffer, 2003; Michailova and Husted, 2003). On the other hand, we have expanded the theory by developing arguments around moderators of these relationships. Our key tenant is that development activities such as employee training have the potential to mitigate the impact of undesirable attitudes via learning mechanisms.

Combined, these insights allow us to draw theoretical implications about the role of attitudes as barriers to the implementation of management strategies, specifically open innovation strategies in the context of manufacturing industries. It is the contention of the article that such insights provide a novel perspective, as the current recommendations for managers tend to praise openness in general, without a proper realization of the difficulties and the potential barriers that may jeopardize its implementation. These insights hence provide a crucial contribution for a deepened understanding of the strategic management of innovation.

From a methodological perspective, our article contributes to the field of innovation management in two important ways. First, it provides data and insights about the working of open innovation for SMEs, an empirical context, which has been less studied in this field (Huizingh, 2011; Spithoven et al., 2013). Second, it presents novel survey-based measures for some of its key concepts, i.e. openness and attitudes to knowledge. In particular, the new approach to measure OI on the basis of a practice perspective was designed to offer a far more comprehensive approach than the general actor-based measures currently available, which rely on inter-organizational relationships as a proxy for openness.

From a practitioner point-of-view, the article calls attention to the attitudinal issues implied in new modes of organizing the

innovation processes. Not all employees are necessarily favorable towards open practices and may resist their implementation. It also specifies some human resource-related practices that can be put in place in order to limit the undesirable effect of the attitudes, depending on which type is most dominant in the organization. In a nutshell, the key message for managers is that negative attitudes to knowledge (the NIH and NSH syndromes) do matter, and that they can be reduced through training programs – either focused on professional development (general or specialized on innovation), or on individualized talent development.

7.2. Limitations and further research

Some limitations of this study are worth mentioning, alongside the opportunities for future research they provide. First, since the analysis is built upon a cross-sectional database, the long-term effects of the attitudes to knowledge could not be investigated. The gathering of longitudinal data in which time lags between variables are present would be an important step for further investigations about the consequences of the NIH and the NSH syndrome. Second, the attitudes have been measured on the basis of the evaluation of managers and not of the employees themselves. Even though the manager's assessment in SMEs can be considered a legitimate and confident proxy, it is not possible to completely rule out the problems of misinterpretation and misjudgment. The application of questionnaires directly to employees in future studies may further enhance our understanding of these attitudes. In addition, as the measures not only of attitudes, but also of open innovation practices have been newly development for this study, further validation of their respective scales is an important step for future research. Third, a single informant approach has been used for the collection of data. Although

a number of statistical tests have been performed to ensure the validity and reliability of the constructs, the potential issue of common method bias cannot be completely rejected. Forth, since employee training has been assessed in the form of three binary variables, future research endeavors should develop more fine-grained measures for employee training to substantiate the relationship between training and innovation. Such variables could include information regarding how many and which employees participate in the training as well as the format (e.g. generic vs. customized programs), the level (e.g. MBA), and the timing of training programs. These measures are also likely to influence employee attitudes. Fifth, the results of the article are naturally limited to the empirical and geographical context in which data was collected. Similar studies in other countries and in other kinds of organizations (i.e. large companies, public sector organizations, service sector) may provide useful and interesting comparative analyses.

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Appendix A

See [Table A1](#).

Table A1

Scale items for the not-invented-here and not-shared-here syndromes.

Construct	Items	Factor loadings	Average variance extracted (AVE)
NIH syndrome	1. Our employees have negative attitudes to applying ideas and technologies from outside.	0.70	0.64
	2. Our employees regard the application of external knowledge as valuable as the application of knowledge generated inside. ^a	0.89	
	3. Our employees have often received and used knowledge from external sources. ^a	0.80	
NSH syndrome	1. Our employees have negative attitudes to having other companies receiving and using our knowledge and technology.	0.67	0.59
	2. Our employees regard external technology commercialization as an equivalent exploitation mode to the application of technologies in own products and services. ^a	0.78	
	3. Our employees have often sold/revealed own knowledge and technologies to other companies. ^a	0.80	
	4. Our employees are positive towards developing new ideas, solutions and technologies for other companies. ^a	0.82	

Note: The items have been measured on a Likert scale from 1 to 7.

^a Reverse coded.

Table B1

Items for employee training.

Variable	Question	Yes (%)
Professional training	Have your employees been offered training aimed at strengthening their professional competences?	79
Training for creativity and innovation	Have your employees been offered training aimed at enhancing their creativity and innovation?	31
Nursing special talents	Have you nursed special talents (by giving them special conditions or training possibilities)?	35

Table C1
Descriptive statistics and correlations among study variables.

Variables	Min	Max	Mean	St. dev.	1	2	3	4	5	6	7	8	9	10	11
1. Inbound OI	0	8	1.11	0.62											
2. Outbound OI	0	4	0.91	0.75	0.251**										
3. NIH syndrome	1	7	2.83	0.90	−0.312**	−0.173**									
4. NSH syndrome	1	7	4.09	1.19	−0.081	−0.260**	0.198**								
5. Professional training	0	1	0.79	0.41	0.266**	0.182**	−0.127*	−0.172**							
6. Training for innovation	0	1	0.31	0.46	0.267**	0.142*	−0.024	−0.045	0.229**						
7. Nursing special talents	0	1	0.35	0.48	0.167**	0.256**	−0.233**	−0.077	0.267**	0.354**					
8. Time for creativity	0	1	0.36	0.48	0.279**	0.245**	−0.176**	−0.071	0.054	0.249**	0.198**				
9. (Ln) Firm age	(0) 1	(5.7) 312	3.38	0.93	0.079	−0.141**	0.095	0.091	0.067	0.063	0.035	0.002			
10. Firm size	0	1	0.51	0.50	0.273**	−0.017	−0.165**	0.069	0.197**	0.166**	0.154**	0.059	0.293**		
11. High-tech industry	0	1	0.09	0.29	0.023	0.108*	−0.140*	−0.078	0.112*	0.043	−0.016	0.049	−0.210**	−0.080	
12. Low-tech industry	0	1	0.26	0.44	−0.101	−0.071	0.170**	0.159**	−0.218**	0.064	−0.027	0.071	0.122*	0.048	−0.187**

* Indicates significant at the 0.05% level (two tailed).

** Indicates significant at the 0.01% level (two tailed).

Appendix B

See Table B1.

Appendix C

See Table C1.

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